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ELEC 547 Assignment: 6

Proposal for the final project

Goal of the project: Experimentation on "**Inpainting using Partial Differential equation Techniques**" (Object removal, scratch removal and text removal)

Paper: "**Region Filling and Object Removal by. Exemplar-Based Image Inpainting. A. Criminisi*, P. Pérez and K. Toyama**" [1]

Introduction

Inpainting is an image interpolation technique used to reconstruct lost or deteriorated parts of an Image. There are three kinds of Image inpainting algorithms

a. Structural inpainting
b. Textural inpainting
c. Combination of both structural and textural inpainting

As the names itself suggests structural and textural inpainting deal with continuity of the structure and the texture of the structure which is being inpainted.

Method presented in the above paper: (combination of both structural and textural inpainting)

Goal: Object Removal

Let 'I 'be the image under consideration and let 'Ω' be the user selected target region, so the source region ϕ is given by ϕ = I - Ω. As the paper mentions the size of the template window ψ chosen is 9x9.

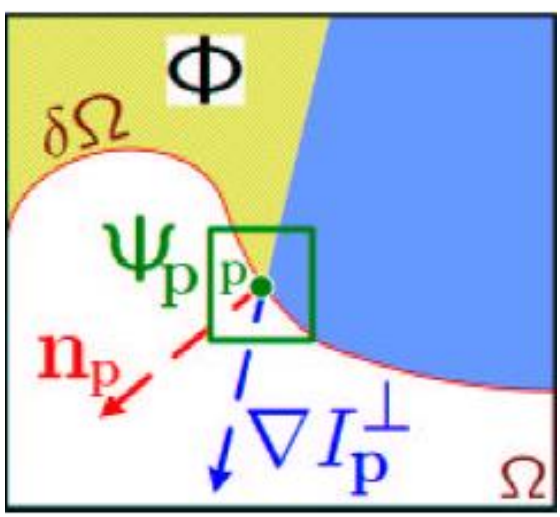

Figure.1 Notation diagram [1]

Once manually selected region is extracted repeat the following steps until all the pixels are filled.

a. Identify the contour of the region to be filled, this contour is called fill front($\delta\Omega^t$)
b. Compute the priorities P(p) $\forall p \in \delta\Omega^t$ using the formula given below

   $P(p) = C(p)\, D(p)$

   where C(p) is confidence term and D(p) is data term

   $$C(p) = \frac{\sum_{q \in \psi_p \cap \bar{\Omega}} C(q)}{|\psi_p|} \quad D(p) = \frac{|\nabla I_p^{\perp} \bullet n_p|}{\alpha}$$

   where $|\psi_p|$ is the area of ψ_p and α is a normalizing factor

   n_p is a unit vector othogonal to $\delta\Omega$ at the point p
c. Find the patch $\psi_{\hat{p}}$ with maximum priority.
d. Look for similar patches in the source region by minimizes SSD in the intensities of the pixels.
e. Fill in the unknown details from the details of the best match patch.
f. Update the confidence term C (p).

What am I going to use?

I am going to use the images presented in the above mentioned paper and few other images from different papers related to object removal using inpainting, to obtain my results and compare them to the state of the art results.

What am I going to implement?

I am going to implement the above mentioned paper completely from scratch (object removal, scratch removal and text removal).

State-of-the art:

The state of the art results in "**object removal based on inpainting**" were provided by Criminisi et.al [1] which is the paper I have chosen. I will compare the results of my algorithm with results in the paper. I will also compare my results with commercial solutions like "**photo wipe**" [2].

Learning outcomes:

- Applications of Partial differential equations in image processing.
- I work with blood vessel data, with this project I can get a better understanding on inpainting and how I can use it on my data.

References:

1. Region Filling and Object Removal by. Exemplar-Based Image Inpainting. A. Criminisi*, P. Pérez and K. Toyama
2. http://www.hanovsolutions.com/?prod=PhotoWipe